SOVEREIGN INFRASTRUCTURE

KEYSTONE SOVEREIGN CORE SERVER + SAIA

Unified Technical Specification

Classification: Confidential — Integrator and Client Eyes Only

Field	Value
Version	1.4
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Prepared By	Polaris IT Solutions
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Audience	Technical Staff, Compliance Auditors, Qualified Legal Counsel
Status	Active — Pre-deployment reference. Parameters will be refined against prototype validation data.

ℹ v1.4 delta from v1.3: (1) Section VII Network Architecture rewritten — three-zone topology, sovereign peripheral zone defined, sovereign peripheral switch specified, office LAN uplink tier progression clarified, Keystone's role as three-zone gateway made explicit. (2) Section IX added — Sovereign MFP Subsystem as first-class infrastructure. (3) Section numbering shifted by one from IX onward. (4) Glossary gains three entries. All other content unchanged from v1.3. Redline against v1.3 for Sections VII and IX specifically.

I. Purpose and Scope

This document is the complete technical and compliance reference for the Sovereign Infrastructure stack — a purpose-built, integrator-deployed system consisting of two components: the Keystone Sovereign Core Server, the trusted root of the sovereign environment, and SAIA, the Sovereign AI Appliance. Together they form a closed, auditable sovereign environment for organizations whose data carries attorney-client privilege, HIPAA obligations, or similar legal weight.

This specification is the definitive pre-deployment reference, version 1.4. It supersedes all prior specifications. Prototype validation and production deployments will inform subsequent revisions. Architecture described herein is correct; specific parameters will be refined against real-world performance data.

1.1 Design Philosophy

- All AI computation is local — no data leaves the sovereign environment without explicit, logged, attorney-approved action
- Sovereign by default — external connectivity is denied unless explicitly permitted and specifically enumerated
- Every privileged action is auditable — ingestion, query, anonymization, and export events are logged immutably
- Entirely open-source — no vendor lock-in, no licensing dependencies, no hidden telemetry
- Hardware-agnostic recovery — the system is recoverable on any compatible x86 platform given drives and passphrase
- Integrator-maintained — not managed by client general IT staff; the integrator relationship is a security control
- Additive, not disruptive — the sovereign stack protects specific privileged data; it does not replace existing email, billing, or calendar infrastructure

1.2 What This Stack Is Not

- Not a cloud service, hybrid cloud solution, or subscription product
- Not a general-purpose workstation or developer machine
- Not a replacement for legal judgment, attorney review, or professional expertise
- Not a system that updates its knowledge from the internet
- Not a product that operates without ongoing integrator engagement
- Not a complete security program — endpoint security, email security, and device management are complementary requirements

II. The Sovereign Stack: Integrated Architecture

The sovereign infrastructure consists of two purpose-built systems and one purpose-specified peripheral subsystem that function as a single integrated environment. Together they establish a closed trust environment for all sensitive data, identity, AI operations, document lifecycle, and physical document production within the firm.

2.1 System Roles

Component	Role
Keystone Sovereign Core Server	The trusted root. Provides identity, key vault, file server, communications, backup, print/scan services, and network gateway between all sovereign zones and the office LAN. Every other system in the sovereign environment depends on Keystone. Keystone does not depend on SAIA. Keystone is a server — the authoritative root of the trust environment — not an appliance.
SAIA Sovereign AI Appliance	The Sovereign AI Appliance. Provides local AI inference, document ingestion, RAG retrieval, anonymization, and the optional attorney-gated cloud polish pathway. Depends on Keystone for identity, secrets, and file storage. Non-functional without Keystone — intentional.
Sovereign MFP Physical ingestion endpoint	The physical ingestion endpoint of the document pipeline. Deployed and hardened by the integrator. Never connected to the office LAN. Communicates exclusively with Keystone via the sovereign peripheral zone. First-class infrastructure — not a peripheral recommendation.

> ℹ Keystone is the Sovereign Core Server. SAIA is the Sovereign AI Appliance. The word 'appliance' applies only to SAIA. Keystone is never described as an appliance — it is the sovereign root, the identity authority, the key custodian, and the network gateway between all sovereign zones.

2.2 Dependency Architecture

Integration Point	Direction	Notes
Identity / SSO (Authentik)	Keystone → SAIA	SAIA authenticates all users through Keystone Authentik. Non-functional if Keystone offline — intentional.
Secrets / Credentials (Vaultwarden)	Keystone → SAIA	SAIA service credentials retrieved from Keystone Vaultwarden at startup. No secrets stored locally on SAIA.
Shared File Server (ZFS/SMB)	Keystone → SAIA	RAG ingestion staging, model storage, and pipeline working files reside on Keystone Data SSD. SAIA mounts over sovereign peripheral zone.
SAIA System Backup	SAIA → Keystone	SAIA OS and vector database drives imaged daily to Keystone saia_backups ZFS dataset.
NTP / Time Authority	Keystone → SAIA	Keystone is the authoritative time source. SAIA syncs to Keystone chrony only.
Print / Scan Services	Keystone → Sovereign peripheral zone	Keystone hosts CUPS print server. MFP authenticates via Authentik LDAP. Print job audit logging to Keystone audit trail.
MFP scan delivery	MFP → Keystone (via sovereign	Scanned documents delivered to scan_quarantine on Keystone. Human review required before promotion to ingestion pipeline.

Integration Point	Direction	Notes
	peripheral switch)	

2.3 The Trust Boundary

The combined system establishes a single, well-defined trust boundary. Inside the boundary: client files, AI inference, attorney work product, internal communications, credentials, physical document production, and all sovereign peripheral devices. Outside the boundary: the public internet, cloud services, AI providers, the office LAN, and any system not under the firm's direct physical control.

The only authorized path for data to exit the trust boundary is the anonymization gate — a strictly enforced, attorney-approved, anonymization-mandatory pathway for optional quality enhancement using an external AI service. The architecture is structurally incapable of routing un-anonymized privileged data to any external service.

2.4 RTO and Availability

Because SAIA has a hard dependency on Keystone for identity and file storage, Keystone's Recovery Time Objective governs the availability of the entire sovereign stack. Keystone's multi-tier backup and sovereign rotation architecture is, in part, the availability architecture for SAIA. A Keystone hardware failure requiring physical intervention will result in SAIA downtime until the integrator performs an on-site service event. No data loss occurs; no unauthorized remote access is possible.

III. Keystone — Sovereign Core Server

Keystone is the Sovereign Core Server — the trusted root of the sovereign environment. Every other system, service, and user depends on Keystone for identity, secrets, file custody, backup integrity, print/scan services, and network gateway between all sovereign zones. Keystone is not a user-facing AI system. It is the security anchor that makes everything else safe to operate.

3.1 Core Services

Service	Component	Description
Identity / SSO	Authentik	Local identity provider. Federates authentication to SAIA and all sovereign appliances. MFA enforced. No cloud identity dependency. Single offboarding action removes all access across all sovereign services simultaneously.
Key Vault	Vaultwarden	Self-hosted Bitwarden-compatible secrets manager. Stores service credentials and operational secrets. Zero cloud dependency. LUKS2 bootstrap keys — master passphrase and Yubikey HMAC keys — are never stored in Vaultwarden. Bootstrap keys are always physical, always offline, always under owner custody.
File Server	ZFS + Samba (SMB)	ZFS-backed SMB file server. Dataset-level permissions. Shadow copy (Previous Versions) enabled. Hosts RAG ingestion staging, model storage, firm files, and SAIA backup targets. Accessed by SAIA over sovereign peripheral zone fiber uplink.
Communications	Matrix Synapse + Element	Sovereign encrypted internal messaging. Replaces Teams or Slack for privileged communications. Behind Authentik SSO.
Intranet / Wiki	Wiki.js or Outline	Internal knowledge base for matter templates and firm procedures. Behind Authentik SSO.
Print / Scan	CUPS + Samba	Network print server for sovereign MFP on sovereign peripheral zone. Authentik LDAP for print authentication. Scan-to-ingest workflow to RAG ingestion dataset via scan_quarantine. Print job audit logging.
Backup	Rescuezilla + ZFS snapshots	Multi-tier backup covering Keystone itself and SAIA. On-machine redundancy, ioSafe physical resilience, ZFS mirror, and sovereign offline rotation to physical safes under owner custody.
Time / NTP	chrony	Authoritative time source for the sovereign environment. SAIA syncs to Keystone only. Monotonic time, hash-chained logs, and administrative clock-change logging provide clock tamper resistance.
Network gateway	Linux routing + nftables	Routes traffic between sovereign peripheral zone, office LAN, and internet. Default-deny outbound. Every permitted flow explicitly enumerated and logged.

3.2 Hardware Specification

3.2.1 Small Firm Deployment (Keystone-1)

Component	Specification	Notes
CPU	Intel i5 10th Gen or newer; i5 7th Gen validated on prototype only	IO-bound workload. Identity, file serving, backup, and network gateway are well within this platform's capability.

Component	Specification	Notes
RAM	32GB DDR4 minimum; 64GB recommended	ZFS ARC benefits from RAM. 32GB is functional minimum. 64GB before deploying Matrix Synapse.
Motherboard	Any LGA1200 or newer with TPM 2.0 header	TPM 2.0 mandatory — onboard or discrete. Required for LUKS2+TPM2 auto-unlock.
Network — sovereign zone	Intel X520-DA2 10GbE SFP+ — uplink to sovereign peripheral switch	Fiber. Galvanic isolation at zone boundary. All SAIA and MFP traffic routes through this interface.
Network — office LAN	Intel i226 2.5GbE — Keystone-1: copper through Furman + optical isolator	Office LAN uplink. Attorneys and paralegals reach sovereign services through this interface only.
Onboard NIC	Disabled in OS	No third active interface. Onboard NIC disabled at OS configuration.
Case	Fractal Design Define R5 — ATX mid-tower, two 5.25" bays	Two 5.25" bays required: IcyDock hot-swap + Blu-ray burner. Sound-dampened for office environment.
PSU	Integral UPS (HDPLEX 500W GaN DC-ATX)	DC-ATX converter drawing from sovereign LiFePO4 battery bank. Full specification in Section 3.3.
Drive Bay	IcyDock 2-Bay Hot-Swap + USB Clone	Mechanical heart of the sovereign rotation system.
Disaster Recovery	ioSafe — floor-bolted	Fire and waterproof. Bolted to floor. Encrypted at rest. Physical disaster recovery layer.

3.2.2 Mid-Sized Firm Deployment (Keystone-2)

Component	Specification	Notes
CPU	Intel i5-13600K (13th Gen) or equivalent	Better concurrency and memory bandwidth. Supports full service stack including Matrix and Wiki.
Motherboard	Z690 or Z790 ATX, DDR4, TPM 2.0	Full PCIe lane support. Mature, stable platform.
RAM	64GB DDR4	ZFS ARC headroom. Supports full service stack comfortably.
Network — sovereign zone	Intel X520-DA2 10GbE SFP+ — uplink to sovereign peripheral switch	Same as Keystone-1.
Network — office LAN	Intel i226 2.5GbE or SFP — fiber direct to office switch	Keystone-2 runs fiber directly to office switch. Galvanic isolation inherent to fiber medium. Optical isolator retired.
Case / PSU / Drive Bay / ioSafe	Same as Keystone-1	Hardware-agnostic recovery architecture identical across tiers.

ℹ Network isolation tier progression: Keystone-1 uses copper to the office switch through Furman + optical isolator. Keystone-2 replaces this with fiber direct to the switch — galvanic isolation inherent, no isolator required. The sovereign peripheral zone uplink uses fiber at all tiers. Keystone-MAX takes sovereign control of the network stack — see 3.2.3.

3.2.3 Large Firm Deployment (Keystone-MAX)

Component	Specification	Notes
CPU	Xeon or Ryzen Pro / Threadripper Pro	Server-grade or workstation-grade processor. ECC RAM mandatory.
RAM	128GB ECC minimum	ECC mandatory at MAX tier. ZFS ARC at scale requires headroom.
Network	Keystone-MAX operates the firm's network stack	At MAX scale, Keystone assumes sovereign control of network infrastructure: routing, VLAN enforcement, DNS, and sovereign Wi-Fi tier. pfSense or OPNsense on sovereign hardware. Full specification deferred — own architecture document.
Storage	High-performance NVMe array	Data SSD tier scales for SAIA backplane throughput saturation.
Services	Full stack + extended services	Identity, Vault, Files, Matrix, Wiki, Print/Scan, extended firm services.

⚠ Keystone-MAX network stack architecture is deferred. Deploying Keystone-MAX without taking sovereign control of the network perimeter places sovereign boundary enforcement in the hands of infrastructure the firm does not fully control. Acceptable at Keystone-1 and Keystone-2 scale; not acceptable at MAX scale. Integrators planning Keystone-MAX deployments must engage the full network architecture specification before deployment begins.

3.2.4 Design Principle: Hardware Agnosticism

Keystone must be recoverable on any available x86 hardware. ZFS pools re-import on any Linux system without reconfiguration. In a motherboard failure scenario, a technician can restore full Keystone functionality on a replacement machine using only the drive complement and the LUKS master passphrase from the offsite safe. No special tools, no vendor support, no proprietary hardware required.

3.3 Power Architecture — Integral UPS

Keystone operates on a sovereign DC power architecture that provides clean, stable power, physical ride-through capability, and elimination of grid noise and transients from the motherboard power rail. The LiFePO4 battery bank is the primary power source; the grid is the charging source. The entire system draws less than one-third of the power architecture's rated capacity under maximum Keystone-1 load.

3.3.1 Component Roles

Component	Specification	Role
Furman SS-6B or equiv	Passive MOV surge suppressor — no firmware, no telemetry	First protection layer. AC input. Network office uplink also passes through at Keystone-1 (copper + optical isolator).
Mean Well HLG-480H-48B	480W, 48V output, fanless passive, 48B variant	AC/DC conversion. Feeds Victron solar input terminal. 48B variant allows output voltage trim.
Analog Power Sensor (APS)	Custom passive build — see 3.3.6. No microcontroller, no firmware.	Three functions: overcurrent protection (15A fuse), binary AC present/absent GPIO signal to motherboard (optocoupler), BMS kickstart resistor (750Ω, 5W).

Component	Specification	Role
Victron SmartSolar MPPT 100/20	Charge controller — 41.5V regulated load output, ≤4.5A charge limit	Manages charge current to battery. Provides regulated load output to system bus.
Schottky Diode	MBR6045WT or VS-60CTQ045-M3, TO-247 dual common-cathode. Bridge Pin 1 + Pin 3 (anodes). Pin 2 tab = cathode = live at system bus voltage.	Passive automatic transfer switch. Forward-biases instantaneously when bus voltage drops below battery voltage on AC loss. No relay, no software, no transfer gap.
LiFePO4 battery — Keystone-1	12Ah 36V, ~432Wh usable	Primary power source on AC loss. Held at ~85% SOC under normal operation — zero daily cycling.
LiFePO4 battery — Pegasus tier	25Ah 36V, ~900Wh usable	Same role, larger capacity for higher sustained load.
HDPLEX 500W GaN	DC-ATX converter, 12–60V input, all ATX rails	Converts system bus DC to ATX power rails. Motherboard never sees AC power or grid noise. Correct spelling: HDPLEX — not HDFLEX.

3.3.2 Operating States

State	Description
Grid-connected	MeanWell holds bus at ~41.5V via Victron load output. Battery anode reverse-biased — battery contributes nothing to load. Battery charges at ≤4.5A via Victron charge output, held at ~85% SOC. Zero daily cycling. Keystone draws ~120W sustained — well under one-third of HDPLEX derated capacity.
Grid-loss	MeanWell collapses. Victron load output sags. Schottky forward-biases passively, instantaneously, relay-free. Battery takes load. APS GPIO goes HIGH. Daemon confirms via secondary network check. Parks mirror HDDs, reducing sustained draw.
Low battery — 33V threshold	Daemon initiates graceful OS shutdown. ZFS pools export cleanly. All services terminate. Threshold adjustable based on observed discharge rate and measured shutdown time at commissioning.
Grid-return	MeanWell recovers. Victron resumes charging at ≤4.5A. Bus returns to 41.5V. Diode reverse-biases. APS GPIO goes LOW. Daemon confirms restoration. Operations resume after battery recovery to target SOC.
BMS bootstrap deadlock	APS kickstart resistor (750Ω, 5W) provides trickle current (~24mA at 48V input, 0.432W dissipation) to battery charge port on AC return. Wakes BMS regardless of discharge port state.

⚠ APS kickstart resistor value: 750Ω, 5W ceramic wirewound. Earlier specification versions listed 100Ω 10W — that value is incorrect and must not be used. At 750Ω worst-case current is 24mA (48V in / 30V battery), dissipating 0.432W. The 100Ω value would produce 480mA worst-case — potentially damaging to the battery BMS. Build all APS units to the 750Ω 5W specification.

3.3.3 Fuse Architecture

⚠ All fuse positions use MC4 inline fuse holders rated 60V DC, 30A continuous, 10 AWG native pigtails. Fuse inserts are 10×38mm cylindrical midget PV DC-rated format only. Automotive blade fuses and MAXI marine blade holders are prohibited — their 32V DC maximum rating is exceeded by every rail in this system.

Rating	Format	Location	Purpose
15A	10×38mm cylindrical PV DC-rated	MeanWell output → APS → Victron PV input	Main input protection, captive to APS assembly
10A	10×38mm cylindrical PV DC-rated	Victron charge output → battery charge lead	Protects 14 AWG charge leads and battery BMS
15A	10×38mm cylindrical PV DC-rated	System bus → HDPLEX input (Keystone-1)	Single branch, all Keystone loads
20A	10×38mm cylindrical PV DC-rated	System bus → HDPLEX input (Pegasus tier)	Higher load tier — 20A provides headroom at battery-floor current levels

3.3.4 Wiring Requirements

Run	Gauge	Notes
Battery discharge → Schottky anode	10 AWG silicone	Main DC trunk
Schottky cathode → System bus	10 AWG silicone	Main DC trunk
System bus → HDPLEX input	10 AWG silicone	Single branch, fuse in line
MeanWell output → APS → Victron PV	10 AWG or 14 AWG silicone	APS pigtail, 15A fuse captive to APS assembly
Victron charge output → battery	14 AWG silicone	10A fuse in line, ≤4.5A charge current
All negatives → Blue Sea busbar	10 AWG silicone	Equal length runs. Single common ground reference. Verify <2 milliohms HDPLEX negative to busbar before first power-on.

3.3.5 Schottky Diode — Thermal Mount Assembly

Open thermal sandwich assembly. Package: TO-247 dual common-cathode (MBR6045WT or VS-60CTQ045-M3). Bridge Anode A (Pin 1) and Anode B (Pin 3) externally. Cathode tab (Pin 2) is live at system bus voltage — never contact aluminum directly. Worst-case dissipation 7.33W at 13.33A and 0.55V Vf.

Layer	Material	Requirement
Upper — heatsink	Anodized aluminum, PEEK-bolted	Active upward heat path
Upper pad	Honeywell PTM7950 or equiv ≥5.0 W/m·K phase-change	Top/heatsink face only. Liquifies above ~45°C — contained by heatsink clamping. Prohibited on bottom face — migrates under pressure.
Diode body	TO-247, cathode tab (Pin 2) live at system bus voltage	Both pads mandatory. Never contact aluminum directly.
Lower pad	Rigid ceramic insulator — alumina, ≥5.0 W/m·K, non-flowing	Bottom/chassis face only. PTM7950 prohibited here.
Lower — chassis	Aluminum bracket or chassis wall	Large thermal mass, primary heat sink

⚠ Hardware: Metal M3 screw with PEEK or ceramic shoulder washer only. Nylon prohibited — creep under torque risks short to live cathode tab. Washer must cover full bore of TO-247 mounting hole — inner ring is tied to live cathode tab at system bus voltage.

3.3.6 Analog Power Sensor

Single passive device performing three functions. No microcontroller. No firmware. Verifiable by inspection against published open hardware schematic. Built x4 units per fleet build session.

Function	Implementation	Values
A — BMS Kickstart Resistor	Series resistor + blocking diode from MeanWell output to battery charge port	R_kick: 750Ω, 5W ceramic wirewound. Blocking diode: 1N4007. Worst-case: 24mA at 48V in / 30V battery. Dissipation: 0.432W.
B — Binary AC Present/Absent Signal	Voltage divider + PC817 optocoupler → motherboard GPIO	R1: 20kΩ 1/4W. R2: 1.5kΩ 1/4W. Forward current: 1.87mA at 41.5V. AC present = GPIO LOW. AC absent = GPIO HIGH. Wire-break = GPIO HIGH — daemon secondary check mandatory before acting.
C — Overcurrent Protection	15A 10×38mm cylindrical PV fuse in MC4 inline holder	Captive on MeanWell pigtail, outside enclosure. 60V DC rated.

Form factor: sealed polycarbonate enclosure (Hammond 1591 series). Three individually-sized cable glands. MeanWell pigtail through dedicated heavy gland. Charging circuit through dedicated gland. GPIO pigtail (24–28 AWG) through tight rubber pass-through with zip-tie strain relief inside wall.

3.3.7 Power Budget — Keystone-1

Load	Watts	Notes
CPU (i5-10600K, Keystone workload)	~65W	IO-bound server workload
Motherboard, RAM	~30W	
OS SSD + Ledger NVMe	~5W	
Data SSD (4TB enterprise)	~5W	
Mirror HDDs ×2 (active)	~10W	Parked on grid loss
Fans, misc	~5W	
Total sustained	~120W	
HDPLEX capacity at 41.5V	~430W	Machine runs at ~28% of derated capacity
Runtime on battery (HDDs parked)	~3.5–4 hours	12Ah 36V pack at ~110W

3.4 Drive Architecture

Drive	Capacity	Role	Filesystem	Encryption
OS SATA SSD	250GB	OS, Authentik, Vaultwarden, databases, keys. Partitioned smaller than physical capacity — bounded image size for rotation media compatibility.	ext4	LUKS2 + TPM2
Ledger NVMe	250GB	Dual role: hourly OS backup target + bootable Rescuezilla recovery environment.	ext4	LUKS2 + TPM2
Data SSD	4TB enterprise SATA (Keystone-1); 8TB (Keystone-2); NVMe array (Keystone-MAX)	ZFS file server — SMB datasets, model storage, RAG staging, SAIA backups	ZFS	LUKS2 + TPM2
Mirror HDD ×2	6TB+ enterprise (Keystone-1); 12TB+ (Keystone-2)	ZFS mirror — active backup pool. Two active, two in sovereign rotation chain.	ZFS mirror	LUKS2 + TPM2
ioSafe	Variable	Fire/waterproof physical disaster recovery backup. Floor-bolted.	ext4 / ZFS	LUKS2 + TPM2

3.5 ZFS Dataset Access Model

Dataset	SMB Access	Notes
rag_ingestion	Ingestion managers (write); SAIA service account (read-only)	RAG pipeline staging. Controlled intake. Every access logged.
firm_files	All authenticated staff (read/write); matter-level subfolder permissions	Privileged matter content. Previous Versions enabled.
saia_backups	SAIA service account only	SAIA OS and vector DB images. Not accessible to end users.
models	SAIA service account (read-only); Integrator (write)	LLM model weights. Updated only by integrator via verified procedure.
scan_quarantine	MFP scan delivery (write); Ingestion coordinator (read/review/promote); SAIA service account (no access)	Physical document intake staging. Human review required before promotion to rag_ingestion. Prevents physical prompt injection via external documents.
infrastructure	Integrator only	System configs, update packages, audit logs, build documentation.

3.6 Keystone Software Stack

Layer	Component	Notes
OS	Ubuntu 26.04 LTS	5-year LTS. Security updates integrator-controlled. No automatic updates.
Identity / SSO	Authentik	Local IdP. MFA enforced. No cloud dependency.
Key Vault	Vaultwarden	Self-hosted Bitwarden-compatible vault. Operational secrets only — never LUKS bootstrap keys.

Layer	Component	Notes
File Server	ZFS + Samba (SMB)	Dataset-level permissions. Shadow copy. Fiber uplink to sovereign peripheral zone.
Print / Scan	CUPS + Samba	Network print server. Authentik LDAP auth. Scan-to-ingest pipeline via scan_quarantine.
Communications	Matrix Synapse + Element	Sovereign encrypted messaging.
Intranet / Wiki	Wiki.js or Outline	Internal knowledge base. Behind Authentik SSO.
Containers	Docker + Compose	All services containerized.
Reverse Proxy	Nginx	TLS termination. Routes through Authentik.
Backup Engine	Rescuezilla + ZFS native	Block-level imaging + ZFS snapshots.
Archival	M-DISC Blu-ray	Write-once archival for models, closed matter snapshots, OS archives.
Time / NTP	chrony	Authoritative time source for sovereign environment.
Network gateway	Linux routing + nftables	Routes between sovereign peripheral zone, office LAN, and internet. Default-deny outbound.

IV. SAIA — Sovereign AI Appliance

SAIA is a locally-deployed AI inference engine. It processes documents, answers questions, and assists with legal research and drafting using AI models that run entirely on hardware physically located in the firm's facility. No query, no document, and no response traverses a public network in normal operation.

SAIA is an architecture, not a product. The models it runs are open-source and auditable. The software stack is open-source and auditable. There are no black-box components, no accounts with third-party AI providers, and no data leaving the building without explicit attorney authorization and anonymization.

4.1 Hardware Specification

4.1.1 Reference Build (SAIA-1)

Component	Specification	Notes
CPU	Intel Core i7-13700K (Raptor Lake, 16 cores / 24 threads, LGA1700)	Sufficient for parallel preprocessing, tokenization, and service orchestration alongside GPU inference.
Motherboard	ASUS PRIME Z790M-PLUS D4, LGA1700, Micro-ATX, DDR4, PCIe 5.0	TPM 2.0 required. Full PCIe 5.0 and 4.0 lane support.
RAM	64GB DDR4 2666MHz — matched or mixed kit, JEDEC default only. No XMP.	Sufficient for LLM runtime, RAG pipeline, OS, and containerized services concurrently.
CPU Cooler	Thermalright SI-100 Black — 100mm top-flow, 6×6mm heat pipes	100mm height — fits SOEYi U320 135mm clearance with 35mm headroom.
CPU Power Limits	PL1: 125W / PL2: 150W / Tau: 10 seconds. Configure in BIOS before any inference workload.	Required spec item. i7-13700K unconstrained can exceed 250W.
GPU	NVIDIA RTX Pro 4000 Blackwell, 24GB GDDR7 ECC, 5th gen Tensor Cores, FP4 support	24GB VRAM supports all reference models at full multi-user concurrency. FP4 validation is a required commissioning step.
Case	SOEYi U320 Phoenix, Micro-ATX cube, 4 carry handles	Conference table deployment. Compact, professional, portable.
PSU	SFX 650W+ 80+ Gold	Headroom for workstation GPU and full system load.
OS NVMe	Enterprise NVMe with hardware Power Loss Protection — Samsung PM9A1 or Intel P41 Plus, 512GB–1TB	PLP mandatory. SAIA has no Integral UPS — abrupt power loss is a real scenario.
Vector DB NVMe	Samsung 980 Pro 2TB, PCIe 4.0	Dedicated Qdrant vector database storage.
Scratch NVMe	Standard NVMe 1TB+	Pipeline working files, model cache, temp inference buffers.
Network	Onboard 2.5GbE copper (sovereign peripheral switch); all other interfaces disabled	Single active network interface. One wire, one destination.
Power	Furman SS-6B or equiv (passive MOV) + CyberPower pure sinewave UPS	Pure sinewave UPS required — confirm model explicitly before purchasing.

ℹ Model weights and RAG ingestion staging reside on the Keystone Data SSD and are accessed over the sovereign peripheral zone. This is a security architecture decision: model weights on a GPU-attached system are more easily exfiltrated; keeping them on Keystone means they are governed by Keystone's dataset permissions and backup chain.

4.1.2 Scaled Deployment Configurations

Configuration	GPU	VRAM	Recommended Use
SAIA-1 (Reference)	1× NVIDIA RTX Pro 4000 Blackwell 24GB GDDR7 ECC	24GB	1–5 concurrent users; small firm deployment.
SAIA-2 (Dual)	2× RTX Pro 4000 Blackwell or equiv — load-balanced, not tensor-parallel. Each card holds a full model instance.	48GB total	5–20 concurrent users; mid-size firm. Primary card anchors inference pool; secondary handles overflow sessions and RAG ingestion.
SAIA-Pro	1× RTX 6000 Ada 48GB or RTX 6000 Pro 96GB	48–96GB	20+ concurrent users; large models; high-volume document processing.
SAIA-Enterprise	1× RTX 6000 Pro 96GB workstation class + expansion	96GB+	Enterprise; multiple practice groups; parallel model serving.

ℹ The dual-card SAIA-2 configuration uses workload segmentation, not tensor parallelism. Two discrete inference engines, each holding a full model instance, load-balanced across users. 70W TDP workstation cards enable this at combined 140W GPU draw in a standard chassis.

4.2 SAIA Software Stack

Layer	Component	Notes
OS	Ubuntu 26.04 LTS	Hardened baseline. Integrator-controlled updates. No automatic updates.
Identity / SSO	Authentik (via Keystone)	SAIA hosts no local IdP. Non-functional if Keystone offline — intentional.
Secrets	Vaultwarden (via Keystone)	All service credentials retrieved from Keystone Vaultwarden at startup.
Inference Runtime	vLLM	High-throughput LLM serving. OpenAI-compatible API. Use AWQ or native FP4 weight formats — not GGUF.
Language Model	Gemma 3 27B AWQ (reference)	Strong reasoning and instruction-following. Fits reference VRAM. Mistral 24B and Qwen2.5 32B are validated alternatives.
Vector Database	Qdrant	Local deployment. No telemetry. Hybrid sparse+dense search.
OCR Pipeline	PaddleOCR + Tesseract	PaddleOCR for complex layouts. Tesseract as auditable fallback.
RAG Orchestration	FastAPI + LlamaIndex	Clean API / retrieval separation.
Anonymization	spaCy NER + custom rules	Local NER. Mandatory gate for any external transmission.
User Interface	OpenWebUI	Multi-user chat interface. No external dependencies.
Containers	Docker + Compose	All services containerized.

Layer	Component	Notes
Reverse Proxy	Nginx	HTTPS termination. Access control. Request logging.
Cloud Polish (optional)	Claude API (Anthropic)	Narrow, audited, anonymization-gated only. Disabled by default. Attorney approval required per use.
GPU Management	nvidia-smi + NVIDIA drivers	Monitor power draw, temperature, VRAM utilization.

4.3 VRAM Architecture and Model Guidance

Scenario	Model	Format	VRAM Est.	Concurrent Users (24GB)
Reference deployment	Gemma 3 27B	AWQ	~14–16GB	3–5
Higher quality	Gemma 3 27B	FP4 (Blackwell native)	~7–8GB	6–10 — validate at commissioning
Smaller / faster	Mistral 24B	AWQ	~12–14GB	4–6
Alternative	Qwen2.5 32B	AWQ	~16–18GB	3–4
Demo / fast response	Gemma 3 12B	AWQ	~6–8GB	8–12

ℹ FP4 validation is a required commissioning step on all Blackwell deployments. FP4 roughly halves VRAM requirement at modest quality cost — if validated, enables larger models or significantly higher concurrency. Test FP4 vs AWQ baseline throughput at commissioning and document tokens/second for both before production use.

V. Document Lifecycle

5.1 Ingestion Pipeline

Documents enter the sovereign trust boundary through two paths: network upload to the RAG ingestion staging share, or physical scan via the sovereign MFP to the scan_quarantine staging share. The scan_quarantine path requires human review and explicit promotion before the document reaches the ingestion pipeline — this prevents physical prompt injection via externally-originated documents arriving via MFP scan.

Stage	Action	Notes
1 — Intake	Document placed in rag_ingestion staging share (network) or scan_quarantine (physical scan)	Physical scans held in scan_quarantine pending ingestion coordinator review.
2 — Human review (physical scans only)	Ingestion coordinator reviews scan_quarantine content before promotion	Mandatory gate. Prevents physical prompt injection. SAIA service account has no access to scan_quarantine.
3 — Preprocessing	Text extraction (PDF, DOCX, TXT, RTF) or OCR (scanned images). PaddleOCR primary, Tesseract fallback.	PaddleOCR handles complex layouts. Tesseract fallback triggered on low-confidence output.
4 — Chunking	Text split into overlapping chunks with metadata (source, page, timestamp)	Overlap preserves context across chunk boundaries.
5 — Embedding	Chunks converted to vector embeddings via local embedding model	No external embedding API. All computation local.
6 — Indexing	Embeddings stored in Qdrant on SAIA Vector DB NVMe alongside source chunk	Hybrid sparse+dense indexing for retrieval quality.
7 — Audit log	Ingestion event logged: document name, hash, timestamp, operator, chunk count	Immutable. Part of the Keystone audit trail.

5.2 RAG Query Pipeline

Stage	Action	Notes
1 — User query	Attorney submits query via OpenWebUI (Authentik-authenticated)	All queries logged: user, timestamp, query text, session ID.
2 — Retrieval	Query embedded; nearest chunks retrieved from Qdrant; hybrid sparse+dense search	Source citations included in retrieval results.
3 — Context assembly	Retrieved chunks assembled into prompt context with source metadata	Context window managed per model spec.
4 — Inference	Prompt submitted to vLLM; model generates response	All inference local. No external API call in normal operation.
5 — Response delivery	Response delivered to user with source citations	Citations display document name, page, and date.

Stage	Action	Notes
6 — Audit log	Query and response logged: user, timestamp, query, response summary, sources cited	Immutable. Available to compliance review.

5.3 Cloud Polish Gate

The cloud polish pathway is disabled by default. When enabled, it is the only authorized path for any data to exit the trust boundary. The pathway is not a general internet connection — it is a narrow, attorney-gated, anonymization-mandatory channel to a single allowlisted external AI endpoint.

Gate	Requirement
Attorney authorization	Explicit per-use approval by a named attorney. Logged with timestamp and authorizing user identity.
Anonymization	All PII, entity names, matter identifiers, and privileged markers stripped by the local spaCy NER pipeline before any content leaves the system. Deanonymization key retained locally for re-integration of polished response.
Endpoint allowlist	Single allowlisted endpoint only. Default: Anthropic Claude API. No other external destination permitted.
BAA requirement	If PHI is possible in the corpus, a Business Associate Agreement with the cloud provider is required before the feature is enabled. Non-negotiable.
Audit log	Every cloud polish invocation logged: authorizing attorney, timestamp, anonymized content hash, response hash, deanonymization key reference.

⚠ The deanonymization key is stored encrypted on SAIA OS NVMe. Loss of the key renders anonymized documents permanently anonymized — the polished response cannot be reintegrated. The key is backed up as part of the SAIA backup chain. Treat the deanonymization key as sovereign data equivalent to Vaultwarden secrets.

5.4 Institutional Knowledge Accumulation

The vector database accumulates the firm's institutional knowledge over time. Matter documents, research, correspondence, and templates indexed against each other create a corpus of increasing value. This corpus is a sovereign asset — it lives on hardware the firm controls, is backed up under the firm's sovereign rotation chain, and is archived to M-DISC at matter close. It cannot be extracted, sold, or discontinued by any third party.

VI. Backup Architecture

6.1 Sovereign Rotation System

Keystone's backup architecture is built around a monthly rotation of encrypted drive clones through a chain of physical locations under owner custody. The rotation chain provides protection against hardware failure, ransomware, site disaster, and any form of remote attack — including attacks against the integrator. The integrator never holds a copy of the firm's data.

Tier	Media	Location	Frequency	Encryption
On-machine — Ledger NVMe	250GB NVMe — dual boot + hourly backup	Inside Keystone	Hourly OS image	LUKS2 + TPM2
On-machine — ZFS mirror	HDD mirror pair (active)	Inside Keystone	Continuous ZFS snapshots	LUKS2 + TPM2
Physical resilience — ioSafe	Fire/waterproof enclosure	Floor-bolted at site	Daily rsync	LUKS2 + TPM2
Sovereign rotation — onsite safe	HDD clone (from mirror rotation chain)	Firm's onsite safe	Monthly clone swap	LUKS2 + TPM2
Sovereign rotation — offsite safe	HDD clone (from mirror rotation chain)	Attorney's personal safe or firm offsite	Monthly clone swap	LUKS2 + TPM2
Long-term archival — M-DISC	Blu-ray M-DISC write-once	Multiple locations	At matter close; annually	LUKS2 at rest; M-DISC inherently immutable
SAIA backup	SAIA OS + vector DB daily image	Keystone saia_backups ZFS dataset	Daily	LUKS2 + TPM2

6.2 Monthly Rotation Choreography

The monthly sovereign rotation is performed by the integrator or a trained designated staff member. The choreography is procedural and physical — drives move between Keystone, the onsite safe, and the offsite safe in a defined sequence that ensures the backup chain is never broken.

Step	Action
1	Retrieve Drive A from onsite safe. Retrieve Drive B from offsite safe (or bring Drive C from Keystone).
2	At Keystone: hot-swap Drive A into IcyDock. Initiate clone via USB clone button. Wait for completion indicator.
3	Verify clone integrity: mount clone image, verify SMART data, confirm file count against source.
4	Label clone with date and rotation sequence number. Update rotation log.
5	Move Drive A (freshly cloned) to offsite safe. Move Drive B to onsite safe. Return Drive C to Keystone mirror pool.
6	Confirm all rotation log entries signed. File rotation record in infrastructure dataset.

6.3 Quarterly Integrity Verification

Every quarter, the integrator performs a full integrity verification: mounts the offsite rotation drive, verifies LUKS2 decryption, confirms ZFS pool health, and performs a test restore of a sample file to confirm the recovery path is functional. Results documented in the infrastructure dataset.

6.4 M-DISC Archival Policy

- Model weights archived to M-DISC when a new model version is deployed — provides a verifiable record of which model was in use at any given time
- Closed matter vector DB snapshots archived to M-DISC at matter close — preserves the matter's AI-assisted research trail for professional responsibility purposes
- OS archive burned to M-DISC annually — preserves a known-good system state for long-term recovery
- M-DISC copies stored at minimum two physical locations — onsite fireproof storage and offsite

VII. Network Architecture

Keystone is the network gateway for the entire sovereign environment. It sits at the intersection of three distinct network zones simultaneously: the sovereign peripheral zone (SAIA, MFP, sovereign peripherals), the office LAN (attorney and paralegal workstations, VPN remote users), and the internet (controlled outbound only). Every byte that crosses a zone boundary passes through Keystone and is subject to its access controls, firewall rules, and audit logging.

This is not incidental to Keystone's role — it is central to it. The Define R5 case, the Integral UPS, the dual-NIC configuration, and the fiber uplink architecture are all expressions of the same requirement: Keystone must be the machine in the breach, continuously available, handling three network zones without becoming the weak point that undermines the sovereignty it exists to provide.

> ◆ The sovereign peripheral switch is the galvanic isolation point for the entire sovereign peripheral zone. One fiber uplink to Keystone. Everything behind it — SAIA, MFP, future sovereign peripherals — connects via copper. Nothing that touches a downlink port of the sovereign peripheral switch is ever a non-sovereign device. This rule is physical, not configured.

7.1 Three-Zone Topology

Zone	Contents	Keystone Interface	Isolation Method
Sovereign peripheral zone	SAIA, MFP(s), sovereign peripheral switch, future sovereign peripherals	X520-DA2 10GbE SFP+ fiber — uplink to sovereign peripheral switch	Fiber uplink provides galvanic isolation at zone boundary. All devices in zone are integrator-deployed sovereign hardware. No office LAN path exists or is permitted.
Office LAN	Attorney and paralegal workstations, VPN remote users, office switch and router	Intel i226 2.5GbE — Keystone-1: copper through Furman + optical isolator; Keystone-2+: fiber direct to office switch	VLAN isolation at network infrastructure layer. All access to sovereign services mediated by Nginx and Authentik. No workstation has any direct path to SAIA.
Internet	Cloud polish endpoint (optional), integrator-controlled update sources, firm SIEM (optional)	Through office LAN uplink — not a separate interface	Default-deny outbound. Host firewall enforced. Every permitted outbound flow explicitly enumerated. OS and GPU telemetry blocked at host firewall.

7.2 Sovereign Peripheral Zone

The sovereign peripheral zone is a defined network segment containing only integrator-deployed sovereign hardware. It has a single uplink — fiber to Keystone — and no other connection to any network. Devices in this zone communicate via copper through the sovereign peripheral switch. The fiber uplink to Keystone provides inherent galvanic isolation at the zone boundary.

Device	Connection	Role in Zone
Sovereign peripheral switch	Fiber uplink to Keystone X520-DA2	Zone boundary device. Managed — sovereign fabric only. No office LAN ports.
SAIA	Copper to sovereign peripheral switch	All Keystone dependency traffic (identity, secrets, file access, NTP, backup) routes through switch to Keystone.
Sovereign MFP	Copper to sovereign peripheral switch	Physical document ingestion endpoint. Scan delivery to Keystone scan_quarantine via SMB. Print jobs from Keystone CUPS.
Future sovereign peripherals	Copper to sovereign peripheral switch	Additional scanners, physical intake hardware, other integrator-deployed sovereign devices. Never added to the office LAN.

⚠ The sovereign peripheral switch must never have a non-sovereign device on any downlink port. This is a physical topology constraint, not a configuration policy. Enforce by physical access control to the switch, not by ACL alone.

7.3 Sovereign Peripheral Switch Specification

Requirement	Specification	Rationale
SFP uplink port	Minimum 1GbE SFP — 10GbE SFP+ preferred to match Keystone X520-DA2	Fiber uplink to Keystone. Galvanic isolation inherent to optical medium.
Copper downlink ports	Minimum 4× GbE copper — 8× preferred	SAIA, MFP(s), and future sovereign peripherals.
Managed switch	VLAN support, management access restriction	Management access restricted to sovereign fabric only. No remote management from office LAN.
Power conditioning	Furman or equiv passive MOV on switch power circuit	Protects sovereign switch fabric from conducted transients originating at MFP or building power.
Physical placement	Server room or secured IT enclosure alongside Keystone	Physical access control enforced.

7.4 Office LAN Uplink — Tier Progression

Tier	Office LAN Interface	Isolation Method	Notes
Keystone-1	Intel i226 2.5GbE — copper Ethernet	Furman passive MOV + optical isolator in cable run	Galvanic isolation achieved via optical isolator. Adequate for Keystone-1 deployment profile.
Keystone-2	Intel i226 2.5GbE or SFP — fiber direct to office switch	Galvanic isolation inherent to fiber medium. Optical isolator retired.	Full fiber deployment. Furman stays on AC side — its optical isolator role on the network side is retired.
Keystone-MAX	Sovereign network stack — Keystone manages the	Keystone IS the network perimeter	Full specification deferred — own architecture document.

Tier	Office LAN Interface	Isolation Method	Notes
	firm's routing, switching, and VLAN infrastructure		

7.5 Network Segmentation

Zone	Can reach SAIA?	Can reach Keystone?	Method / Notes
Attorney/Paralegal VLAN	Yes (HTTPS)	Yes (HTTPS + SMB)	Authentik authentication required. Primary access path.
Remote users (VPN)	Yes	Yes	Firm VPN + Authentik MFA required.
Admin endpoint	Yes (SSH only)	Yes (SSH only)	Key-based SSH on management interface only. Documented recovery path.
General office LAN	No	No	VLAN isolation. Enforced at network infrastructure layer.
Guest / Wi-Fi	No	No	No exceptions.
Internet (inbound)	No	No	No port forwarding. Neither system reachable from public internet.
Sovereign peripheral zone	Direct (copper via switch)	Direct (fiber uplink)	Single fiber uplink is the only gateway between sovereign peripheral zone and all other networks.
MFP (print/scan)	No	Yes (CUPS/SMB via sovereign switch)	Authenticates via Keystone Authentik LDAP. Scan output to scan_quarantine for human review.

7.6 Outbound Traffic Policy

Default-deny outbound on both systems. Every permitted outbound flow is explicitly defined.

Destination	Appliance	Permitted?	Condition
Cloud polish endpoint (Anthropic API)	SAIA via Keystone	Conditional	Feature enabled; anonymization gate passed; attorney approval; every request logged.
Software / model update source	Both	Conditional	Integrator-initiated only; scheduled maintenance window; hash verification required.
Firm SIEM	Both	Conditional	Firm-configured; pull-based preferred; explicit setup required.
OS telemetry (Ubuntu)	Both	No	Blocked at host firewall.
GPU telemetry (NVIDIA)	SAIA	No	Blocked at host firewall.
Any other destination	Both	No	Default deny. Blocked and alerted.

7.7 Internal Service Mesh (SAIA)

Service	Container	Exposed To	Network Access
User Interface	OpenWebUI	Authenticated users via Nginx	Internal only
Inference Engine	vLLM	FastAPI orchestration only	Internal service mesh
RAG Orchestration	FastAPI + LlamaIndex	OpenWebUI; Authentik-validated	Internal only
Vector Database	Qdrant	FastAPI only	Internal only
OCR Pipeline	PaddleOCR + Tesseract	Ingestion pipeline only	Internal only
Anonymization	spaCy service	Ingestion pipeline + cloud polish gate	Internal only
Reverse Proxy	Nginx	External VLAN (HTTPS 443 only)	Single ingress point

VIII. Physical Document Production

8.1 MFP Integration Architecture

The sovereign MFP is the physical intake point for the document pipeline. It is deployed and hardened by the integrator, placed on the sovereign peripheral zone exclusively, and has no path to the office LAN or any external network. Full MFP specification is in Section IX. This section covers the print and scan workflows that connect the MFP to the broader document pipeline.

8.2 Scan-to-Ingest Workflow

Stage	Action
1 — Scan at MFP	Staff authenticates at MFP panel (PIN-based or LDAP card). Scans document to Keystone scan_quarantine share via SMB.
2 — Watcher detection	Keystone scan watcher service detects new file in scan_quarantine. Appends metadata: timestamp, source device, operator identity (if PIN auth configured).
3 — Ingestion coordinator review	Designated ingestion coordinator reviews scan_quarantine content before promotion. This gate is mandatory — prevents physical prompt injection via externally-originated documents.
4 — Promotion to ingestion pipeline	Coordinator promotes document to rag_ingestion staging share. Document enters ingestion pipeline (Section V).
5 — Audit log	Scan event, review action, and promotion logged to Keystone audit trail. Chain of custody established from physical document to sovereign document store.

8.3 Print Workflow

Print jobs from sovereign workstations route through Keystone CUPS — not sent directly from workstation to MFP. Keystone is the control point for all print traffic. Every print job is logged: authenticated user, document name, timestamp, page count. Print job logs are part of the Keystone immutable audit trail.

8.4 Bates Stamping and Discovery Production

Document production for discovery is handled as a governed workflow: documents retrieved from the sovereign file server, Bates stamps applied by designated staff using integrator-approved tooling, output written to a controlled production dataset. The production dataset is subject to the same access controls and audit logging as all sovereign datasets. Chain of custody from original document to production set is documented in the audit trail.

IX. Sovereign MFP Subsystem

The sovereign MFP is a first-class component of the Sovereign Infrastructure stack. It is the physical ingestion endpoint of the document pipeline — the point at which paper documents enter the sovereign trust boundary. Its configuration, network posture, and physical placement must be treated with the same rigor as Keystone and SAIA.

The sovereign MFP is distinct from any leased office copier. The leased copier remains in place and on contract. The sovereign MFP is an integrator-deployed, integrator-hardened device that is never connected to the office LAN, communicates exclusively with Keystone via the sovereign peripheral zone, and has no path to any manufacturer cloud or external service.

9.1 Manufacturer Selection — Brother Industries

Brother Industries is the approved manufacturer for sovereign MFP deployments. The selection is based on sovereign deployment criteria:

- SMB/CIFS push scanning without cloud account or registration requirement
- Full network configuration via embedded web interface and BRAdmin — no manufacturer cloud dependency
- Disableable WiFi, WiFi Direct, and USB host/device ports — firmware-level, not policy-level
- Native CUPS compatibility for Keystone print server integration
- Clean firmware update path via BRAdmin or embedded web interface — no manufacturer cloud required
- Family-controlled Japanese manufacturer — no foreign acquisition history, no cloud-extraction business model
- Consolidated high-yield toner and drum consumable family across monochrome and color lines — single SKU stocking for multi-device and multi-site deployments

> ℹ The consumable consolidation across Brother's MFP line is a meaningful operational benefit in law firm deployments. A single toner/drum SKU covering both monochrome and color devices at a site simplifies procurement, reduces stockout risk, and eliminates the technician error of installing the wrong consumable. It is a considered selection criterion.

9.2 Deployment Tiers

Tier 1 — Sovereign Scan Node (Default)

Brother MFC-L6915DW

Attribute	Specification
Print	Monochrome laser
Scan	Color CIS, 1200dpi optical
ADF	Duplex ADF, 80-sheet capacity
Duplex print	Yes
Network	Gigabit Ethernet — WiFi disabled at deployment
Street price	~$500–600
Consumable family	TN-X / DR-X high-yield — shared with Tier 2

The L6915DW is the primary recommendation for all deployments. Color laser print is not a law firm workflow requirement — color scan is, and this device delivers it at full optical resolution. Monochrome print is fast, reliable, and economical. This is the default sovereign scan node.

Tier 2 — Redundant + Color Print Capable

Brother MFC-L6915DW + Brother MFC-L8905CDW (paired)

Attribute	MFC-L6915DW	MFC-L8905CDW
Print	Monochrome laser	Color laser
Scan	Color	Color
ADF	Duplex, 80-sheet	Duplex, 70-sheet
Street price	~$500–600	~$750–850
Consumable family	TN-X / DR-X	TN-X / DR-X (shared)

The paired configuration adds color laser print, doubled scan throughput, and ADF redundancy simultaneously. Both devices share the same consumable SKU. Recommended for mid-size firms or any firm where document ingestion is a primary daily workflow.

> ℹ A single L8905CDW may be deployed as the sole Tier 2 device where color print is required but volume does not justify a pair. The crossover point at which a single Tier 3 device is preferable to a Tier 2 redundant pair is higher than it appears — two devices provide redundancy, workflow separation, and consumable simplicity that a single high-volume device cannot match organizationally.

Tier 3 — High-Volume (Site Assessment Required)

Brother MFC-9000 series. For deployments with 25+ users in sustained heavy utilization. Requires site assessment before specification. Not a default escalation path.

Companion Device — Light Duty / Attorney Desk

Brother MFC-L3780CDW. For small offices, single-attorney deployments, or desk-side use where a full workgroup MFP is not warranted. Color laser, DADF, duplex, full sovereign scan capability. Street price ~$450–500. Integrates with the same Keystone scan pipeline.

9.3 Network Architecture

> ◆ The sovereign MFP is never connected to the office LAN. This is a physical topology constraint, not a configuration policy. The device cannot reach the office LAN because it is not connected to it.

Topology	Description
Standard deployment (Tier 1 and Tier 2)	MFP(s) → Sovereign peripheral switch [copper downlinks] → Keystone [fiber uplink — galvanic isolation]

Topology	Description
Minimal deployment	MFP → Sovereign peripheral switch → Keystone. The sovereign peripheral switch is present in all deployments — it is the galvanic isolation point and the sovereign peripheral zone boundary.

Fiber is used for the uplink between the sovereign peripheral switch and Keystone. An MFP is a high-current switching power supply connected to building electrical infrastructure. A copper Ethernet path from the MFP environment to Keystone is a conducted noise and surge path into the most sensitive appliance in the stack. The fiber uplink breaks this path completely at the zone boundary.

9.4 MFP Hardening Checklist

All sovereign MFP deployments require completion of the following checklist before the device is placed in service. The integrator documents completion in the deployment record.

Network hardening

- ☐ Wired Ethernet configured — static IP or DHCP reservation from Keystone
- ☐ WiFi radio disabled (Network Configuration > WLAN > Disabled)
- ☐ WiFi Direct disabled — verify independently from WiFi radio setting
- ☐ All scan destinations cleared
- ☐ Scan-to-SMB configured to Keystone scan staging share only
- ☐ No other scan destinations present
- ☐ Test scan to staging share completed and verified
- ☐ No outbound internet path available — verified by network topology, not by ACL

USB hardening

- ☐ USB host ports disabled (Function Lock / Security settings)
- ☐ USB device port disabled
- ☐ No USB connection paths present or active

> ⚠ USB is a copper path with power pins. The device communicates exclusively via its wired Ethernet interface. No USB paths are permitted under any operational circumstances — including break-glass scenarios.

Access control

- ☐ Default admin credentials changed
- ☐ Admin password stored in Keystone Vaultwarden
- ☐ Web admin interface password-protected
- ☐ Physical panel PIN lock enabled
- ☐ Panel PIN stored in Keystone Vaultwarden
- ☐ BRAdmin remote management access restricted to sovereign fabric only

Firmware

- ☐ Firmware updated to current release at deployment
- ☐ Firmware version documented in deployment record
- ☐ Firmware update source: Brother official only via BRAdmin or embedded web interface
- ☐ Firmware review scheduled as part of integrator maintenance cadence

Registration

- ☐ No cloud registration performed
- ☐ No manufacturer account created
- ☐ No registration prompts accepted or deferred — registration disabled where configurable

9.5 Keystone Integration

The Keystone scan watcher service monitors scan_quarantine and processes incoming files: detects new scan jobs; appends metadata including timestamp, source device, and operator identity; notifies ingestion coordinator for review; clears processed files on confirmed promotion. Print jobs from sovereign workstations route through Keystone CUPS — Keystone is the control point for all sovereign peripheral traffic. Where PIN-based scan authentication is configured, the scan watcher logs operator identity against each ingested document, providing a chain-of-custody record from physical document to sovereign document store.

9.6 Client-Facing Specification Language

> *Sovereign document scanning is provided by an integrator-deployed Brother MFC workgroup laser MFP, selected and configured to the deployment tier. The device operates exclusively on the sovereign peripheral network and has no connection to the office LAN or to any external network or manufacturer cloud service. WiFi, WiFi Direct, and USB connectivity are disabled at deployment. No cloud registration is performed. Administrative credentials are managed by Keystone's key vault. Scan jobs are delivered directly to Keystone's secure staging share via SMB over the sovereign fiber-isolated network fabric, where they are held for human review before entering the document ingestion pipeline.*

X. Security Architecture

10.1 Encryption at Rest

Every storage device on both systems is encrypted with LUKS2. TPM2 provides automatic unlock on legitimate hardware — the system boots without a passphrase prompt when the TPM2 seal matches. Yubikey HMAC-SHA1 provides a hardware-bound secondary unlock slot. The master passphrase, sealed in an envelope, is the break-glass recovery path.

Unlock Method	When Used	Custody
TPM2 auto-unlock	Normal boot on legitimate hardware	Hardware-bound. Cannot be transferred to another TPM.
Yubikey A or B	TPM2 unavailable; hardware replacement; integrator service event	One Yubikey per sovereign rotation kit. Two keys enrolled — either unlocks. Hardware-bound, non-exportable.
Master passphrase	Yubikeys unavailable; full recovery scenario	Sealed in envelope. Stored in offsite safe under owner custody. Never stored in Vaultwarden.

10.2 Motherboard Failure Recovery

On motherboard failure: replace motherboard (same or compatible platform), boot from Ledger NVMe into Rescuezilla, unseal LUKS2 with Yubikey or master passphrase, restore OS drive from most recent image, re-enroll new TPM2, return to service. The ZFS Data SSD and mirror HDDs reimport without reconfiguration. Total data loss: zero. Downtime: one integrator on-site service event.

10.3 Defense-in-Depth

Layer	Control
Physical	Keystone and SAIA in locked server room or secured enclosure. ioSafe floor-bolted. Sovereign peripheral switch in same secured enclosure. Physical access log maintained.
Network	Three-zone topology with Keystone as sole gateway. Default-deny outbound. No inbound from internet. VLAN isolation for office LAN. Sovereign peripheral zone physically isolated — fiber uplink only.
Host	LUKS2+TPM2 encryption. AppArmor profiles on service containers. Minimal installed packages. No GUI. SSH key-based only. Automatic updates disabled — integrator-controlled patch cycle.
Application	Authentik SSO with MFA enforced. Nginx reverse proxy — no direct service exposure. Per-dataset SMB permissions. Vaultwarden access restricted to management endpoints.
Data	ZFS checksumming and scrub schedule. LUKS2 encryption at rest. Immutable audit log. Anonymization gate enforced for any external transmission.
Operational	Integrator-only update authority. Sovereign rotation chain under owner custody. No integrator data custody. Quarterly integrity verification. Staff training requirement.

10.4 Threat Model

Threat	Control
Ransomware / malware on office network	Sovereign peripheral zone physically isolated — no office LAN path. SAIA has one wire, one destination. Keystone VLAN isolation. Sovereign rotation chain is airgapped and offline.
Compromise of a cloud AI provider	No data sent to any cloud provider in normal operation. Cloud polish is optional, disabled by default, anonymization-mandatory, and attorney-gated.
Malicious document injection (physical)	scan_quarantine gate — human review mandatory before physical scans enter ingestion pipeline. Prevents physical prompt injection via externally-originated documents.
Insider threat — attorney or staff	Authentik audit trail. Dataset-level permissions. No staff access to infrastructure dataset. Print and scan audit logs. Key vault access restricted to management endpoints.
Integrator compromise or malfeasance	Integrator never holds encryption keys. Sovereign rotation drives are owner-custody at all times. Master passphrase sealed and offsite. Owner can recover without integrator involvement given drives and passphrase.
Hardware failure	Multi-tier backup. Ledger NVMe hourly. ZFS mirror continuous. ioSafe daily. Sovereign rotation monthly. Zero data loss on any single hardware failure.
Physical site disaster	ioSafe fire/waterproof on-site. Sovereign rotation drives at offsite safe. M-DISC archives at multiple locations.
Remote network attack	No inbound ports from internet. No port forwarding. Perimeter firewall default-deny inbound.

10.5 Host Hardening Baseline

- Ubuntu 26.04 LTS — no GUI, minimal packages, no unnecessary services
- SSH key-based authentication only — password authentication disabled
- Automatic updates disabled — integrator-controlled patch cycle
- AppArmor profiles enforced on all Docker service containers
- Host firewall (nftables) — default-deny both directions, explicit allow only
- OS and GPU telemetry blocked at host firewall
- Login banner configured — unauthorized access notice
- auditd enabled — system call audit logging
- Fail2ban or equiv — brute force protection on SSH
- BIOS/UEFI hardened: admin password set, USB boot disabled, Secure Boot enforced where compatible

XI. Audit Trail and Accountability

11.1 Event Categories

Category	Events Logged
Identity and access	Login, logout, failed login, MFA event, SSO delegation, password change, account create/disable/delete
Document ingestion	Document name, hash, timestamp, operator, chunk count, ingestion path (network vs physical scan), promotion from scan_quarantine
RAG queries	User, timestamp, query text, session ID, response summary, sources cited, inference duration
Cloud polish	Authorizing attorney, timestamp, anonymized content hash, response hash, deanonymization key reference
File access	SMB access events per dataset — user, timestamp, operation, file path
Print and scan	Print: authenticated user, document name, timestamp, page count. Scan: operator identity, source device, destination, timestamp.
Backup and rotation	Backup job completion, clone integrity check, rotation event, quarterly verification result
System events	Boot, shutdown, service start/stop, LUKS2 unlock method used, clock change (administrative)
Network gateway	Permitted and denied flows across zone boundaries. Every cloud polish request logged at gateway level.
Integrator access	Every integrator SSH session: timestamp, duration, commands executed (auditd)

11.2 Log Integrity Protections

- Logs written to append-only ZFS dataset — no modification or deletion without pool export
- Monotonic timestamps — chrony-disciplined from Keystone NTP authority
- Hash-chained log entries — each entry includes hash of previous entry
- Administrative clock changes logged as explicit audit events — clock manipulation is detectable
- Log dataset not accessible via SMB — accessible only via integrator SSH or Docker service accounts

XII. Update and Maintenance Governance

12.1 Update Procedure

All software updates are integrator-initiated and performed during scheduled maintenance windows. Automatic updates are disabled on both systems. No update occurs without integrator knowledge and action.

Update Type	Procedure
OS security patches	Integrator reviews patch notes. Applies during scheduled window. Verifies service health post-update. Documents in infrastructure dataset.
Docker service updates	Integrator pulls updated image. Verifies hash against published digest. Tests in staging if available. Deploys during scheduled window.
Model updates	New model weights verified against published hash. Previous model archived to M-DISC before replacement. New model deployed and validated at commissioning checklist level before production use.
Firmware (MFP)	Reviewed as part of integrator maintenance cadence. Applied via BRAdmin or embedded web interface — Brother official source only. Version documented in deployment record.
BIOS updates	Integrator discretion. Only when security advisory or stability requirement exists. Not routine.

12.2 Integrator Responsibilities

- Maintain current knowledge of sovereign infrastructure architecture and all deployed component versions
- Perform monthly sovereign rotation procedure or train and verify a designated staff member to perform it
- Perform quarterly integrity verification of backup chain
- Respond to hardware failure within agreed SLA — on-site service event
- Maintain documentation in the infrastructure dataset — current and accurate
- Review and apply security patches on a defined schedule
- Never retain copies of client encryption keys or rotation drives

XIII. Compliance Mapping

13.1 Attorney-Client Privilege

Attorney-client privilege protects confidential communications between attorney and client made for the purpose of legal advice. The sovereign infrastructure is designed to ensure that AI-assisted legal work does not inadvertently waive privilege by routing privileged content through third-party systems without appropriate controls.

Privilege Concern	Sovereign Infrastructure Control
Disclosure to third parties	All AI inference is local. No privileged content transmitted to any third party in normal operation. Cloud polish is anonymization-mandatory, attorney-gated, and logged.
Inadvertent disclosure via cloud AI	Architecture is structurally incapable of routing un-anonymized privileged data to any external service. Anonymization gate is not a policy — it is the only authorized code path.
Metadata exposure	No content, metadata, or query logs transmitted externally. Audit logs remain within the sovereign trust boundary.
Third-party vendor access	No third party has access to the system, its data, or its encryption keys. The integrator's access is audited and bounded.

13.2 ABA Model Rule 1.6 — Confidentiality

Rule 1.6 requires lawyers to make reasonable efforts to prevent the inadvertent or unauthorized disclosure of, or unauthorized access to, information relating to the representation of a client. The sovereign infrastructure addresses this obligation by:

- Eliminating cloud transmission of client data in normal operation
- Encrypting all client data at rest with LUKS2 + TPM2
- Restricting access through Authentik SSO with MFA enforcement
- Maintaining immutable audit trails of all access and operations
- Providing a physically sovereign backup chain under client/attorney custody

13.3 ABA Model Rule 1.1 — Competence

Rule 1.1 requires lawyers to keep abreast of changes in the law and its practice, including the benefits and risks associated with relevant technology. Use of AI in legal practice implicates this rule. The sovereign infrastructure supports competent AI use by:

- Providing source citations for all AI-assisted research — enabling attorney verification
- Running known, auditable open-source models — no black-box outputs
- Requiring attorney approval for any external AI invocation
- Maintaining records of which model was used for any given matter
- Providing integrator training and documentation to support staff competence

13.4 HIPAA

Where the firm handles Protected Health Information (PHI) — personal injury, medical malpractice, workers compensation, or other health-related matters — HIPAA's Security Rule requirements apply to electronic PHI (ePHI). The sovereign infrastructure supports HIPAA compliance through:

- Access controls — Authentik SSO with MFA, dataset-level permissions
- Audit controls — immutable audit trail of all access and operations
- Integrity controls — ZFS checksumming, LUKS2 encryption, hash-chained logs
- Transmission security — no ePHI transmitted externally in normal operation; anonymization-mandatory if cloud polish is used with PHI-possible corpus
- Business Associate Agreement — required from cloud AI provider before cloud polish is enabled for any corpus that may contain PHI

XIV. Scope and Limitations

14.1 What the Stack Validates

- Local AI inference on privileged legal documents is architecturally feasible without cloud dependency
- Attorney-client privilege can be preserved through the full AI-assisted document workflow
- Sovereign backup and rotation provides genuine ransomware and disaster resilience
- The integrator relationship functions as a security control when integrator access is properly bounded and audited

14.2 Known Limitations

Limitation	Notes
AI model knowledge cutoff	Local models have a training cutoff. They do not know about recent case law, regulatory changes, or current events after their training date. Attorney verification is always required for time-sensitive legal research.
RAG hallucination	Retrieval-augmented generation significantly reduces hallucination but does not eliminate it. Source citations must be verified by the attorney. The system is a research assistant, not a legal authority.
Physical security dependency	The sovereign architecture protects data in transit and at rest. It does not protect against an attacker with physical access to the server room who has sufficient time and technical capability. Physical access controls are a prerequisite.
Endpoint security dependency	The architecture protects the sovereign servers. Attorney workstations and their security posture are outside this stack's scope. A compromised workstation can access sovereign services as the authenticated user.
Model bias and errors	AI models can exhibit systematic biases or errors. Legal work product generated with AI assistance requires attorney review before any reliance.
RTO dependency on integrator	Keystone hardware failure requires an integrator on-site service event. RTO is bounded by integrator response time and parts availability, not by the architecture.

14.3 The Human Trust Layer

The sovereign infrastructure provides the technical architecture for safe, privileged AI-assisted legal work. It does not provide judgment. The attorney remains responsible for: verifying AI-assisted research; approving cloud polish invocations; performing or supervising sovereign rotation; training staff; and maintaining the physical security of the server room and rotation drives. Technology reduces risk. It does not eliminate the need for human judgment and professional responsibility.

XV. Deployment Tiers

15.1 Keystone Tier Comparison

Attribute	Keystone-1	Keystone-2	Keystone-MAX
Target	Solo / small firm	Mid-size firm	Large firm / multi-site
CPU	i5 10th Gen or newer	i5-13600K or equiv	Xeon / Ryzen Pro / Threadripper Pro
RAM	32–64GB DDR4	64GB DDR4	128GB+ ECC
Network — sovereign zone	X520-DA2 fiber to sovereign peripheral switch	X520-DA2 fiber to sovereign peripheral switch	X520-DA2 fiber to sovereign peripheral switch
Network — office LAN	Copper + Furman + optical isolator	Fiber direct to office switch	Sovereign control of network stack
Battery	12Ah 36V LiFePO4 (~432Wh)	12Ah 36V LiFePO4 (~432Wh)	Scaled to load
Storage	4TB enterprise SATA data; 6TB+ mirror HDDs	8TB enterprise SATA; 12TB+ mirror HDDs	NVMe array; 20TB+ mirror HDDs
Matrix Synapse	64GB RAM prerequisite	Yes	Yes
Case	Fractal Define R5	Fractal Define R5 or equiv	Site-specific

15.2 SAIA Tier Comparison

Attribute	SAIA-1	SAIA-2	SAIA-Pro	SAIA-Enterprise
GPU	RTX Pro 4000 Blackwell 24GB	2× RTX Pro 4000 Blackwell	RTX 6000 Ada 48GB+	RTX 6000 Pro 96GB+
VRAM	24GB GDDR7 ECC	48GB total	48–96GB	96GB+
Users	1–5 concurrent	5–20 concurrent	20+ concurrent	Enterprise multi-group
Case	SOEYi U320 Phoenix	Site-specific	Site-specific	Site-specific
Cloud polish	Optional, gated	Optional, gated	Optional, gated	Optional, gated

XVI. Organizational Requirements

16.1 Required Policies

Policy	Minimum Requirements
AI Acceptable Use Policy	Permitted use cases; prohibited uses; attorney review requirement; record-keeping obligations; cloud polish authorization procedure
Data Classification Policy	Which matter types are sovereign-eligible; PHI handling; cloud polish eligibility by data classification
Sovereign Rotation Policy	Rotation custodians (primary and backup); rotation schedule; safe access procedures; lost-drive procedure
Incident Response Procedure	Suspected breach; hardware failure; lost rotation drive; integrator escalation contacts
Staff Onboarding / Offboarding	Authentik account provisioning and deprovisioning; single offboarding action removes all sovereign access

16.2 Staff Training Requirements

Role	Training Required
All staff	What the sovereign infrastructure is and is not; how to access SAIA; when attorney review is required; how to report concerns
Attorneys	Cloud polish authorization procedure; source citation verification; professional responsibility obligations for AI-assisted work
Ingestion coordinators	scan_quarantine review procedure; what constitutes a suspicious document; when to escalate
Rotation custodians	Monthly rotation choreography; safe access and custody procedures; quarterly integrity verification steps
Integrator	Full architecture; all deployment and recovery procedures; update governance; audit trail review

XVII. Glossary

Term	Definition
Analog Power Sensor (APS)	Open hardware passive device performing three functions: (A) 750Ω 5W kickstart resistor + 1N4007 blocking diode providing ~24mA trickle to battery charge port on AC return — resolves Victron bootstrap deadlock; (B) 20kΩ/1.5kΩ voltage divider + PC817 optocoupler delivering binary AC present/absent GPIO signal to motherboard; (C) 15A 10×38mm PV fuse overcurrent protection. No microcontroller. No firmware. Verifiable by inspection.
Authentik	Open-source identity provider running on Keystone. Provides SSO, MFA, and user lifecycle management. No cloud dependency. Single offboarding removes all access.
Brother Industries	Approved manufacturer for sovereign MFP deployments. Selected on the basis of sovereign deployment criteria: SMB/CIFS push scanning without cloud dependency, disableable wireless and USB ports, native CUPS compatibility, family-controlled Japanese manufacturer with no cloud-extraction business model, and consolidated consumable family across monochrome and color product lines.
CUPS	Common Unix Printing System. Print server on Keystone. Provides network print services to sovereign peripheral zone. Generates print job audit logs.
Deanonymization key	Substitution map that reverses anonymization. Stored encrypted on SAIA OS NVMe. Loss renders anonymized documents permanently anonymized. Backed up as sovereign data. Treated as equivalent to Vaultwarden secrets.
HDPLEX 500W GaN	DC-ATX power converter. Accepts 12–60V DC input from LiFePO4 battery bus and delivers standard ATX power rails. Correct spelling: HDPLEX — not HDFLEX.
IcyDock	Hot-swap drive bay enclosure with USB clone capability. The mechanical mechanism for Keystone's sovereign rotation system.
Integral UPS	Sovereign DC power architecture built into Keystone. Battery is primary power source; grid is charging source. Transfer on AC loss is passive and instantaneous via Schottky diode. Components: Furman, MeanWell HLG-480H-48B, Victron SmartSolar MPPT 100/20, LiFePO4 battery, Schottky diode (MBR6045WT), HDPLEX 500W GaN, Analog Power Sensor.
ioSafe	Commercially available fire and waterproof external drive enclosure. Keystone's physical disaster recovery backup target. Bolted to floor.
Keystone	The Sovereign Core Server. Trusted root of the sovereign environment. Provides identity, key vault, file server, communications, print/scan, backup, and network gateway. SAIA depends on Keystone; Keystone does not depend on SAIA. Keystone is a server — never an appliance.
Ledger NVMe	250GB NVMe drive. Dual purpose: hourly OS drive backup target and bootable Rescuezilla recovery environment. First tool reached in any recovery scenario.
LiFePO4	Lithium iron phosphate battery chemistry. Flat discharge curve, thermal stability, exceptional cycle life (2000+ cycles at 85% SOC ceiling). Flat curve is critical — HDPLEX receives stable input voltage throughout a power event.
LUKS2	Linux Unified Key Setup version 2. Standard Linux full-disk encryption. All storage on both systems is LUKS2 encrypted.
M-DISC	Millenniata write-once optical media. Verified multi-decade longevity. Cannot be overwritten, reformatted, or remotely wiped. Used for archival of models, closed matter vector DB snapshots, and long-term OS archives.
MFP	Multifunction printer/copier. Sovereign MFP is integrator-deployed and hardened. Placed on sovereign peripheral zone exclusively. Never connected to office LAN. Authenticated via Authentik LDAP through Keystone CUPS. Integrated into scan-to-ingest workflow with mandatory scan_quarantine staging.

Term	Definition
Qdrant	Open-source vector database running on SAIA. No telemetry. Hybrid sparse+dense search. Dedicated NVMe storage.
RAG	Retrieval-Augmented Generation — document retrieval from a vector database combined with language model generation for grounded, source-cited responses.
Rescuezilla	Open-source disk imaging and recovery tool. Boots from the Ledger NVMe. Provides guided recovery for common failure scenarios.
SAIA	Sovereign AI Appliance. Locally-deployed AI inference engine providing RAG, document processing, anonymization, and optional attorney-gated cloud polish. Depends on Keystone for identity and file storage. Cannot function without Keystone.
Sovereign Core Server	Preferred term for Keystone. Used in all client and integrator communications. Keystone is never described as an appliance.
Sovereign peripheral switch	The managed network switch that forms the boundary of the sovereign peripheral zone. Provides fiber SFP uplink to Keystone and copper downlinks to SAIA, MFP(s), and future sovereign peripherals. Management access restricted to sovereign fabric only. Nothing non-sovereign ever touches a downlink port.
Sovereign peripheral zone	The defined network segment containing only integrator-deployed sovereign hardware. Bounded by the sovereign peripheral switch, which provides a single fiber uplink to Keystone and copper downlinks to zone devices. Fiber uplink provides galvanic isolation at the zone boundary. Nothing that connects to a downlink port is ever a non-sovereign device — physical topology constraint, not a configuration policy.
Sovereign rotation	Monthly procedure by which encrypted drive clones are physically rotated between Keystone, onsite safe, and offsite safe under owner custody. Provides airgapped, ransomware-resistant backup sovereignty.
TPM2	Trusted Platform Module version 2. Hardware security chip holding cryptographic keys for LUKS2 auto-unlock. Keys are hardware-bound and cannot be transferred to another TPM.
Vaultwarden	Self-hosted Bitwarden-compatible secrets manager on Keystone. Stores service credentials and operational secrets. Never stores LUKS bootstrap keys — those are always physical, offline, under owner custody.
Victron SmartSolar MPPT 100/20	Charge controller in the Integral UPS. Accepts MeanWell DC output on solar input terminal. Manages charge current to LiFePO4 battery (≤4.5A). Provides regulated 41.5V load output to system bus.
vLLM	High-performance inference engine for large language models. OpenAI-compatible API. Multi-user concurrent access. Deploy with AWQ or FP4 weight formats — not GGUF — to use Tensor Core optimization paths.
VRAM	Video RAM on the GPU. Determines maximum model size and concurrent user capacity on SAIA.
Yubikey	Hardware security key enrolled as LUKS2 secondary unlock slot. Two Yubikeys (A and B) enrolled — either unlocks the system. Hardware-bound; private keys non-exportable. Each travels with its sovereign rotation kit.
ZFS	Zettabyte File System. Enterprise-grade filesystem with checksumming, snapshots, mirroring, and data integrity guarantees. Used on Keystone for file server and backup storage.
ZFS ARC	Adaptive Replacement Cache — ZFS in-memory read cache. 32–64GB is the recommended RAM range for Keystone workloads.

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